Aligned Perspectival Kernel Projections and Bias Reification

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Abstract

This paper develops a multistage, zero-shot Natural Language Inference architecture for large-scale classification using Riemannian tools to formalize semantic geometry. It examines the "geometric fiction of objective neutrality": the reification of a frame-dependent coordinate, metric, and aggregation rule as an unmarked center of political discourse. Ground News provides the principal case of horizontal expansion through publisher-level, exogenous bias metadata. In response, the paper proposes a vertical, content-conditioned and observer-indexed program informed by topos theory and Homotopy Type Theory, without claiming that either formalism automatically supplies an NLP algorithm. The Johnson–Lindenstrauss lemma motivates dimensional adequacy by making the distortion cost of flattening a finite semantic configuration mathematically explicit. The resulting framework treats disagreement as structured evidence while retaining factual and journalistic constraints.

Declaration

I acknowledge the use of Gemini CLI, Codex CLI, and ChatGPT to generate the equation LaTeX, generation of repository code, as well as serving as a sounding board for the machine learning and mathematical operations as the system developed. I confirm that this is entirely my own work and that it has not been submitted for assessment as part of any other programme.

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1 Introduction and Literature Review

1.1 The geometric fiction of objective neutrality

This paper develops a multistage, zero-shot Natural Language Inference (NLI) architecture for large-scale classification in which a Riemannian metric provides a formal language for semantic geometry. The geometric language is not ornamental. It permits disagreement to be represented through locally varying distances and curvature rather than reduced at the outset to a single scalar label. The motivating problem is political, technical, and epistemological at once. The Overton Window names the range of positions that a public at a particular time treats as politically discussable (Mackinac Center for Public Policy, 2019); the boundaries of that range are neither timeless nor independent of the institutions that measure and publicize it. Meanwhile, the V-Dem Institute's 2023 report identifies increasing political polarization in approximately one quarter of countries, about half of them established democracies (V-Dem Institute, 2023). In that setting, the categories through which news and public argument are made intelligible are not secondary bookkeeping. Classification is the "cold-start fuel" driving the artificial-intelligence systems on which contemporary information infrastructures increasingly rely.

The central thesis of this review is that objective neutrality, when represented as an unmarked center of a semantic coordinate system, is a *geometric fiction*. The word *fiction* does not mean that every description is false, that evidence has no force, or that there are no more and less responsible journalistic claims. It means that a coordinate origin, an axis, a metric, and an aggregation rule are products of a representational frame, even when an interface presents them as properties of the represented object itself. A center is always a center with respect to some space of alternatives, some weighting of observations, and some rule of distance. Erasing those dependencies does not remove perspective; it converts a marked perspective into an apparently natural geometry.

This argument rejects two symmetrical reductions. The first is pure relativism: the inference that, because semantic judgment is framed, all interpretations are interchangeable. The second is unity-focused agreement: the inference that responsible classification must

discover or manufacture one canonical label toward which divergent judgments should converge. The proposed alternative is interperspectival. Complex vector embeddings are used not only to plot a general shape of bias but also to analyze its parts as physical properties: distances, directions, local deformations, densities, and paths through a manifold. The eventual methodology therefore follows several vertical tracks. It conditions semantic measurements on declared hypotheses, compares the resulting local geometries, and studies simulated particle movement across the manifold in order to make salient inferences about the perspectival boundaries that define discourse. "Physical" here identifies a mathematical mode of inquiry, not a claim that political meaning is literally a material particle system.

The literature review proceeds from a concrete case to a formal proposal. Ground News is the primary case because it makes a familiar promise particularly visible: it contextualizes coverage by locating news publishers along a left–center–right scale. We describe this as *horizontal expansion*, since information about an article is enlarged by attaching metadata drawn from a source-level classification system. We then contrast it with a *vertical* approach in which the same content is repeatedly interrogated under explicit observer or hypothesis conditions and the relations among those conditioned representations become the object of analysis. The contrast is not between a commercially impure system and a view from nowhere. Prompts, hypotheses, model weights, sampling procedures, and metrics are also interventions. The difference is that the vertical program keeps those interventions indexed and available for comparison instead of allowing one of them to disappear into a global center.

This distinction provides a bridge among work that is too often separated into technical and philosophical domains. Research in perspectivist NLP establishes that annotator disagreement can be evidence rather than error. Work in journalism and democratic theory establishes that pluralism cannot simply be equated with civility or consensus. Kuhn, Goodman, Bowker and Star, and Wittgenstein show why measurement cannot be understood as the passive registration of a ready-made order. Topos theory and Homotopy Type Theory (HoTT), approached here as formal inspirations and a research

program rather than as turnkey machine-learning algorithms, suggest ways to speak rigorously about context-indexed objects, compatible local descriptions, and paths between interpretations. Finally, the Johnson–Lindenstrauss lemma supplies a quantitative distortion budget for asking how much dimension is needed to preserve a finite relational configuration. Together these resources sharpen the paper's governing question: how can a classifier retain factual constraint while refusing to reify one projection of discourse as its neutral, Platonic form?

1.2 Ground News and horizontal expansion

Ground News is a private news-aggregation platform organized around reducing the effects of media bias. Its interface groups reporting on a topic and displays, among other contextual fields, the distribution of sources assigned to positions on a political bias scale. Its advertising consequently presents politically plural information streams as a means of improving contemporary knowledge and critical judgment. Bias is easily read within this product grammar as a negatively coded deviation from neutrality: left and right appear as departures from a perceived middle, while the middle occupies a visually and linguistically privileged location. Ground News's own methodology is more careful than that simplified reading. Its rating-system explanation states that a "Center" rating does not by itself mean that a source is balanced, unbiased, or neutral (Ground News, n.d.). That disclaimer matters, yet it does not dissolve the geometric issue. A centered, one-dimensional taxonomy continues to structure the display even when the platform denies that the center guarantees epistemic virtue.

Operationally, the displayed bias rating is a publication-level classification rather than an article-level semantic judgment. Ground News explains that its ratings draw on three external assessment organizations: AllSides, Ad Fontes Media, and Media Bias/Fact Check (Ground News, n.d.). AllSides in turn uses, among other procedures, multiperson blind bias surveys to estimate source leanings (AllSides, n.d.). The practical benefit is plain. A reader can quickly inspect whether a story is being covered by publishers that the rating system has classified differently, discover outlets outside an habitual infor-

mation stream, and ask why coverage volume or framing varies. Nothing in the present critique requires denying that such a product can increase awareness or sometimes reduce polarization.

The relevant issue is the unit and provenance of the rating. Let $a \in \mathcal{A}$ be an article, let $p(a) \in \mathcal{P}$ denote its publisher, and let

$$m \colon \mathcal{P} \longrightarrow \mathcal{M}$$

be a metadata map assembled through a prior rating process. A horizontally expanded record has the form

$$H(a) = \big(a, m(p(a))\big) \in \mathcal{A} \times \mathcal{M}, \tag{1}$$

and a displayed bias category or score may be idealized as

$$b_{\mathrm{hor}}(a) = \psi(m(p(a)))\,, \tag{2}$$

where ψ translates publisher metadata into an interface category. Equation (1) is an *extrinsic build-up*: the record becomes wider because a new field is appended, not because a semantic property has been inferred from the linguistic signal of a. The publisher identifier and its established profile exist before and outside the article's particular claims. In this precise sense, the explanatory metadata lies on a separate informational channel from the content it contextualizes.

Calling the system horizontal should not be inflated into the inaccurate claim that Ground News performs no processing of content anywhere in its aggregation service. Aggregation, story clustering, headline comparison, and other product functions may require content processing. The narrower, defensible claim is that the political bias badge shown for an item is inherited from a source-level, exogenous classification rather than produced as a direct semantic judgment of that article. This qualification preserves what is valuable in the student's original objection. The user encounters article a, but the politically salient coordinate is principally a property assigned to $p(a)$. A source can

publish heterogeneous genres, authors, arguments, and departures from its editorial norm; nevertheless, those article-level variations are not the unit represented by the displayed publisher rating.

Horizontal measurement is therefore not uniquely a Ground News flaw. Ground News is important because it signals a broad methodological tendency in anti-bias and recommender systems: uncertainty about content is managed by importing a prior classificatory framework, after which the framework is treated as context for the object rather than as another object requiring context. The architecture connects data points by lines drawn at a global, source-classification level. This can be useful, but it also raises the question posed by Bowker and Star (1999): do classification systems merely describe a social reality, or do their categories, standards, and residual cases actively organize it? Once a platform formalizes a middle, its taxonomy influences what readers can readily perceive as a difference, an extreme, an omission, or a bridge.

The proposed vertical alternative changes the unit of analysis from the publisher profile to a set of content-conditioned relations. Let x denote the article text or a structured segment of it, let $o \in \mathcal{O}$ index an observer condition, and let h_o be the explicit NLI hypothesis or framing intervention associated with that condition. The basic representation is

$$z_o(x) = E_\theta(x; h_o) \in \mathbb{R}^d, \tag{3}$$

where E_θ is a model with parameters θ. The vertical record is not x plus a settled label about its source. It is the family

$$V(x) = \{z_o(x),\ r_o(x, h_o),\ g_o\}_{o \in \mathcal{O}}\,, \tag{4}$$

where r_o records an observer-conditioned semantic relation and g_o may specify an observer-indexed metric or local metric tensor. The analytical signal lies both in each section and in the transformations among them. Verticality thus means repeated, declared, content-conditioned probing of one input and examination of depth across perspectives. It does not mean access to an uncontaminated essence of the text.

1.3 Measurement as an active constituent of worlds

The journalistic debate over objectivity acquires direct philosophical force once classification is rendered as geometry. Kuhn's account of scientific revolutions denies that observation can be cleanly separated from the paradigms that determine which problems, instruments, variables, and operations count as scientifically significant (Kuhn, 1962). Measurement on this account is not an arbitrary imposition, but neither is it a transparent window. A paradigm trains attention, supplies exemplars, and establishes the comparisons through which a datum becomes a datum. Kuhn should not be read as totally rejecting reality or as claiming that recalcitrant evidence never constrains inquiry. His more useful lesson here is that there is no framework-free observation language in which rival classificatory systems can be compared without remainder.

Goodman's theory of worldmaking intensifies the constitutive point. Worlds are made through symbolic practices of composition and decomposition, weighting, ordering, deletion and supplementation, and deformation (Goodman, 1978). The stars offer the student's most vivid image. Luminous bodies may exist independently of a diagram, but a constellation is not simply waiting in the sky as a naturally bounded individual. It is achieved by selecting points, drawing relations, naming a figure, transmitting a convention, and learning to see according to it. Measurement similarly makes sets, saliences, and continuities. A scale does not create the material it encounters ex nihilo, yet it helps constitute the object *as measured*: as hot or cold relative to calibration, as left or right relative to an axis, or as central relative to a distribution.

The original draft describes Goodman's position as a kind of "mereological nihilism." That phrase can be retained only as a strong interpretation of worldmaking, not as a settled description of Goodman's view. His pluralist or irrealist thesis is not the simple claim that there are no objects, parts, or constraints. It is that objects and kinds are available under versions whose modes of composition are not uniquely dictated by a bare, versionless world. The constellation image accordingly avoids two errors. It does not say that the stars vanish when a culture redraws the figure; and it does not say that every grouping is equally serviceable for navigation, astronomy, ritual, or politics. Constitution

is not unlimited license.

Bowker and Star carry this point into infrastructures. Categories distribute visibility, labor, resources, and accountability; they often become most powerful when their historical choices disappear into routine operation (Bowker and Star, 1999). The apparently descriptive middle of a bias scale therefore has practical consequences. It establishes adjacency, measures deviation, and collects heterogeneous cases under a shared name. These operations can stabilize coordination, but their stability should not be confused with neutrality. A classificatory framework both reveals and produces structure: it discloses patterns within its terms while helping make those terms socially consequential.

Wittgenstein offers a related limit on the fantasy of a complete, unpositioned representation. In the *Tractatus*, he writes that "the subject does not belong to the world: rather, it is a limit of the world" (Wittgenstein, 1922, §5.632). The proposition need not be imported wholesale as a theory of NLP. Its value here is diagnostic. The standpoint from which a world is projected cannot be treated straightforwardly as just another already-neutral object inside the completed projection. Every map has a relation to its maker, coordinate choice, and purpose. When those conditions are suppressed, the map's shadow can be mistaken for the thing's total form.

This is the geometric fiction in its mature form. Observers cast light and encounter shadows. Principal Component Analysis makes the image literal by projecting a high-dimensional configuration onto directions selected for variance. A political scale performs an analogous, though not identical, act of selection. Projection is unavoidable in any finite interface, so the objection cannot be that Ground News projects while some alternative does not. The objection arises after projection, when one historically produced shadow is reified as a Platonic view of the entire dimension. Participants in an AllSides blind bias survey each cast their own light and shadow; their responses can disclose an approximate shape of bias over time. Yet an aggregate source rating hides many of the paths by which that approximation was made. Once it is attached to an article and then displayed on a single axis, the observations undergo a double collapse: first from a distribution of upstream judgments into publisher metadata, and second from that metadata into a

one-dimensional approximation.

Kuhn, Goodman, Bowker and Star, and Wittgenstein thereby support a concatenatable but limited inference. Middle positions are not intrinsically neutral points. They are effects of a representational frame applied to an object under specific social and mathematical operations. This conclusion does not invalidate measurement; it makes measurements answerable for the worlds they help constitute. Nor does it prohibit a global description. It requires that any proposed global description establish how local perspectives are related, which invariances it preserves, and why its gluing rule deserves authority.

1.4 Disagreement as semantic evidence

Perspectivist NLP supplies an empirical and methodological counterpart to this philosophy. Its central reversal is to treat disagreement as a signal in its own right rather than automatically as noise around a latent gold label. Frenda et al. (2024) survey approaches that preserve annotator identities, perspectives, or label distributions instead of collapsing subjective judgments into a single canonical target. Interpretive diversity is then not merely a nuisance to be averaged away; it may disclose ambiguity, contested concepts, differential experience, or multiple coherent readings. Xu and Jurgens (2026) extend this orientation by arguing that disagreement is widespread across NLP tasks and requires modeling and evaluation practices designed for it, rather than metrics that treat consensus as the sole endpoint.

Natural Language Inference is especially useful because it localizes judgment in a relation. A familiar NLI task asks whether a premise entails, contradicts, or is neutral with respect to a hypothesis. It might appear that careful training and precise instructions should remove variance. Yet Pavlick and Kwiatkowski (2019) show that inherent disagreements remain in human textual inferences even under controlled annotation conditions. The same linguistic input can support systematically different judgments because annotators import distinct assumptions about quantification, typicality, reference, world knowledge, and pragmatic context. These outputs need not be random. They can be

internally coherent relative to different interpretive frames.

More detailed annotation rules can raise agreement scores by narrowing the assumptions annotators are permitted to use, but increased agreement does not automatically imply increased annotation quality. CrowdTruth begins from this problem and from the observation that forcing workers to mimic a presumed expert answer can conceal the semantic variation the task was meant to measure (Aroyo and Welty, 2015). In the student's apt phrase, an over-specified protocol may amount to *perfuming the agreement scores*. Workers appear to converge because the examiner's assumptions have been installed in the instructions. The resulting dataset may be more regular while becoming less candid about why reasonable interpreters initially differed.

CrowdTruth is consequently a foundation for the system proposed in this paper. It rejects the "golden middle" as the automatic truth of human annotation and jointly models contributions from workers, media units, and annotation choices. That move demonstrates that disagreement can be structured and that the quality of a label cannot always be separated from the pattern of workers who supply it. It also opens an important path beyond label aggregation. A histogram of votes retains more information than a majority label, but it still does not necessarily represent the geometry among interpretations: which judgments are close, which assumptions transform one into another, where paths are discontinuous, and which minority view remains stable under perturbation.

The present project therefore attempts to begin where CrowdTruth ends. Aroyo and Welty's provocative tendency to associate truth outside mathematics with agreement or consensus is powerful as a critique of naively objective gold labels, but it risks preserving the very dichotomy the present system seeks to escape: objective physics on one side and subjective sociology on the other. NLI provides another route. Semantic relations can be modeled as artifacts of the encounter among an input x, a structured hypothesis h_o, a model E_θ, and an observer-indexed measurement rule. The output is neither a feature discovered without mediation nor merely a free opinion. It is a reproducible relation produced under declared conditions.

There is also a curatorial limit within disagreement-aware pipelines. CrowdTruth-

style analyses may distinguish disagreement attributed to ambiguous media, poor task design, ontological mismatch, spam, or low-quality annotation. Such distinctions can be indispensable for data collection. Nevertheless, as soon as some sources of variance are classified as useful signal and others as noise, disagreement has already passed through a heuristic sorting method. The preserved signal is curated, not raw. A response diagnosed as task failure may instead reveal that the ontology offered by the task does not fit the text; an apparent outlier may express a minority frame absent from the instructions. The point is not that filtering is illegitimate. It is that filtering must itself become an observable operation in the semantic geometry.

This recasts agreement as something to be modeled after, rather than before, its existence is established. If several observers yield nearby embeddings or mutually transportable judgments, the local coherence is evidence. If their results diverge along stable directions, that structure is also evidence. If an interpretation disappears under small prompt changes, its instability is evidence of another kind. The system thereby remains critical of forced consensus without presuming that plurality is formless. Constraints arise from the text, from the structured relations of NLI, from repeatability, and from the geometry of transformations among frames. Perspectivism becomes a disciplined account of conditioned invariance, not a declaration that all labels are equal.

1.5 Facts, civility, pluralism, and democratic conflict

Journalistic ethics places a necessary boundary on any enthusiasm for interpretive plurality. The International Federation of Journalists' Global Charter requires fidelity to facts that are known and makes factual responsibility a basic professional obligation (International Federation of Journalists, 2019). Observer dependence cannot license fabrication, erase the difference between a supported and unsupported report, or turn every politically useful assertion into an equally valid world. A perspectival classifier should expose how facts are selected, related, emphasized, and inferred while continuing to distinguish those operations from inventing evidence. The rejection of a neutral center is therefore not a rejection of factual constraint.

The harder question is whether consensus-oriented presentation is identical with pluralism. Ground News's left–right comparison can plausibly help readers encounter unfamiliar sources and see coverage gaps. At the same time, the spectrum introduces axial bias and depends on a methodology that most users cannot reconstruct from the display alone. More fundamentally, bias-centered aggregation tends to produce a managed, civility-oriented discourse. Rival claims are arranged around a stable midpoint, and political maturity can come to look like equal distance from the rated poles. That arrangement is a substantive moral and political choice, not a merely technical convenience.

Reiheld's analysis of civility clarifies the risk. Civility may serve the functional purpose of making society work, or it may embody respect for persons; these purposes can overlap but are not identical (Reiheld, 2013). An exceptionless demand for civility can turn into a simplistic functional relativism when the maintenance of polite interaction takes precedence over contesting beliefs that threaten another person's viable form of life. Berlin's value pluralism treats human differences as constitutive rather than accidental and draws liberal consequences of toleration from that plurality (Berlin, 1998). Pye likewise links pluralistic democracy, particularly where rival moral conceptions meet, to demanding practices of civility (Pye, 1999). Yet respect for the person who holds a belief does not require representing every consequence of that belief as harmless, nor does empathy eliminate the need for judgment.

This is the tension Reiheld names as exceptionless civility: reluctance to impose any value beyond a unifying imperative to keep discourse functioning. The apparent neutrality of the middle can reproduce that reluctance geometrically. Harm, exclusion, institutional power, and the practical incompatibility of forms of life may be compressed into symmetrical distances from center. A platform may then offer plural sources while retaining a single normative topology in which disagreement is valuable primarily because it can be harmonized.

Democratic theory gives reason to resist that reduction. Conflict is not necessarily corrosion of a political system; it can be a constitutive process that requires institutions and controlled procedures rather than elimination (Swanstrom and Weissmann, 2005).

Mouffe (2000) argues that a consensus pursued as an end in itself may neutralize antagonism and weaken democracy's capacity to stage real alternatives. A genuinely plural information system must therefore do more than place adversaries into adjacent bins. It should make visible what the disagreement is about, how perspectives transform the semantic field, where factual constraints hold, and which conflicts cannot be represented as mere symmetrical deviation.

The project occupies a deliberately narrow passage between these concerns. Against relativism, it retains the IFJ's factual constraint and seeks structural stability across repeated measurements. Against forced unity, it refuses to define truth as whatever survives a global vote. Against exceptionless civility, it treats conflict as an analyzable feature of democratic discourse rather than a contaminant to be removed. Its aim is not a perfectly agnostic machine—which would repeat the fiction of a view from nowhere—but an accountable system whose observer conditions and geometric consequences can be inspected.

1.6 Observer-indexed geometry: from a center to relations

The formal consequence of this literature is that semantic geometry must be indexed. In Equation (3), h_o can encode a structured political hypothesis, interpretive role, or intervention; the index o may denote a human observer, an annotator population, a prompt-defined frame, or a reproducible analytical condition. These are not interchangeable social entities, and the methodology must record which meaning is intended. For fixed x, the family $\{z_o(x)\}_{o\in\mathcal{O}}$ makes observer dependence explicit. A relation such as entailment can likewise be written $r_o(x, h_o)$, and distance can be indexed as

$$d_o(x, x') = d_{g_o}(z_o(x), z_o(x'))\,, \tag{5}$$

where g_o is a Riemannian metric tensor or a specified local approximation. Different observer conditions may preserve some relations while bending others. Localized curvature is precisely the resource that allows semantic disagreement to change across regions

rather than being forced into a uniform global scale.

Even the most familiar definition of a center reveals its dependence on a frame. Let $(\mathcal{M}, g)$ be a metric or Riemannian space, let nonnegative weights satisfy $\sum_{o\in\mathcal{O}} w_o = 1$, and suppose a minimizer exists. The weighted Fréchet mean of the observer-conditioned representations is

$$c_{w,g}(x) = \arg\min_{y\in\mathcal{M}} \sum_{o\in\mathcal{O}} w_o \, d_g(y, z_o(x))^2 \,. \tag{6}$$

The notation makes visible what a bare word like "center" conceals. The result depends on the observer family $\mathcal{O}$, the weights w, the representation map E_θ, the hypotheses h_o, and the metric g. On a curved manifold the minimizer can also be nonunique outside suitable convex neighborhoods. A center may be a useful estimator, but it is not an unmarked origin delivered by discourse itself.

In a Euclidean chart, the information lost by reporting only the mean can be stated exactly. With $\bar{z} = \sum_o w_o z_o$,

$$\sum_o w_o \|z_o - \bar{z}\|_2^2 = \frac{1}{2} \sum_{o,o'} w_o w_{o'} \|z_o - z_{o'}\|_2^2. \tag{7}$$

The left-hand side is weighted dispersion about the mean; the right-hand side is average pairwise disagreement. Replacing the family by $\bar{z}$ discards exactly the dispersion encoded by those pairwise relations. Aggregation can therefore retain a location while destroying the shape that made the location meaningful. This is the algebraic version of the double-collapse objection: the shadow's centroid survives, while the competing lights and the distances among their shadows disappear.

A one-dimensional political projection makes the loss more direct. Let $u \in \mathbb{R}^d$ be a unit vector and define

$$P_u(z) = u^\mathsf{T} z. \tag{8}$$

If $z - z' \in \ker P_u$, then $P_u(z) = P_u(z')$ even when $\|z - z'\|_2 > 0$. For $d > 1$, the kernel has dimension $d-1$: many distinct movements are invisible along the selected axis. Two articles can share a left–right coordinate while differing sharply in institutional trust, economic assumptions, affect, national identification, evidential style, or the hypotheses

under which an NLI system treats their claims as entailed. The projection is not false because it is partial. It becomes misleading when its null directions are interpreted as an absence of difference.

The proposed vertical analysis accordingly treats dimensions as relations discovered under interventions, not as a fixed list of political essences. It asks which changes in h_o move $z_o(x)$, which preserve neighborhoods, where local metric tensors differ, and whether paths between interpretations encounter sharp boundaries. Simulated particle movement provides a later methodological means of probing these properties: path cost, attraction, resistance, and local curvature can make perspectival boundaries salient. The present introduction claims only the conceptual warrant for that strategy. It does not assume that the learned manifold is a complete ontology of politics or that a neural embedding transparently reproduces human understanding.

1.7 Topos-theoretic and HoTT-informed verticality

Topos theory gives a disciplined vocabulary for the transition from one global coordinate system to context-indexed descriptions. Let $\mathcal{C}$ be a category of observer contexts whose morphisms represent admissible refinements, comparisons, or changes of frame. A presheaf

$$\mathcal{E} : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{Set} \quad \text{or} \quad \mathbf{Vect} \tag{9}$$

assigns a representation space $\mathcal{E}(o)$ to each context and a restriction or transport map to each context morphism. An article can then have a local section $s_o(x) \in \mathcal{E}(o)$ rather than one contextless coordinate. The maps record how much of a representation survives when the context is refined or translated. This is the formal analogue of refusing to detach a shadow from the light that produced it.

The word *sheaf* adds a nontrivial requirement. One must specify a coverage relation on $\mathcal{C}$, identify compatible local sections on overlaps, and state a gluing condition under which those sections determine a global section (Mac Lane and Moerdijk, 1992). The present paper uses this language as an organizing research program. It does not claim that merely naming observers creates a sheaf, nor that topoi automatically yield an NLP

algorithm. More importantly, observer dependence does not prove that no global section can exist. A defensible neutrality claim would require more than the existence of some aggregate section: it would require a canonically distinguished section invariant under the observer transformations judged relevant to the task. The vertical architecture does not assume such a privileged section in advance; it investigates compatibility and obstruction empirically.

HoTT offers a complementary, not identical, inspiration. In its foundational interpretation, types behave like spaces, terms like points, and identity proofs like paths; identities between identities can in turn be represented by higher paths (Univalent Foundations Program, 2013). Applied cautiously to semantic modeling, this suggests that the relation between two interpretations need not be exhausted by declaring them equal or unequal. There may be multiple paths from one to another, corresponding to different sequences of hypothesis changes, and those paths may themselves agree or differ. The relevant object is thus not only a set of labels but a structured space of transformations.

This analogy is philosophically attractive because it resists two flattenings at once. It need not force locally distinct representations into literal identity, and it need not treat every difference as incomparable. Transport along a declared path can preserve some structure while changing other structure. That is an apt formal ideal for interperspectival analysis. Nevertheless, the paper does not claim that its code implements genuine HoTT, that learned vector homotopies are automatically identity types, or that HoTT establishes the intrinsic observer dependence of language. Those would be category mistakes. Topos theory and HoTT discipline the questions asked of the architecture: What is local? What is transportable? Which diagrams commute? Which paths are distinct? When can local evidence be glued? The answers must still come from specified models, data, and tests.

The vertical proposal can now be stated schematically:

$$x \longmapsto \{z_o(x)\}_{o\in\mathcal{O}} \longmapsto \{\Phi(z_o(x))\}_{o\in\mathcal{O}} \longmapsto \big\{d_o,\, g_o,\, \text{transport and topological diagnostics}\big\}, \tag{10}$$

where Φ is a declared feature map or projection. This pipeline remains conditioned by

θ, by the choice of hypotheses, and by the geometry imposed or learned downstream. Its advantage is not metaphysical purity. Its advantage is traceability: the frame is a variable in the analysis, and agreement or disagreement appears as a relation among local measurements rather than as a source label appended before the article is examined.

1.8 Dimensional adequacy and the Johnson–Lindenstrauss lemma

The light-and-shadow metaphor raises a quantitative question. If all observation involves projection, how much dimension is required to retain the relations of interest? The Johnson–Lindenstrauss (JL) lemma provides a standard answer for finite Euclidean point configurations. Let $0 < \varepsilon < 1$ and let $X = \{x_1, \dots, x_m\} \subset \mathbb{R}^d$. A useful explicit form states that when

$$k \geq 4 \left(\frac{\varepsilon^2}{2} - \frac{\varepsilon^3}{3} \right)^{-1} \log m, \tag{11}$$

there exists a map $f \colon \mathbb{R}^d \to \mathbb{R}^k$ such that for all i, j,

$$(1 - \varepsilon)\|x_i - x_j\|_2^2 \leq \|f(x_i) - f(x_j)\|_2^2 \leq (1 + \varepsilon)\|x_i - x_j\|_2^2. \tag{12}$$

Thus $k = O(\varepsilon^{-2} \log m)$ dimensions suffice to preserve all pairwise squared distances within controlled multiplicative distortion (Johnson and Lindenstrauss, 1984; Dasgupta and Gupta, 2003).

The lemma is significant here as a distortion budget, not as a proof that meaning possesses one true Euclidean dimension. It says that a rich but finite configuration can often be compressed dramatically while retaining its pairwise geometry; it also says that the target dimension must scale with the number of points and the fidelity demanded if one wants a uniform worst-case guarantee. A generic one-dimensional center offers no comparable promise for many observer-conditioned semantic states. Equation (8) identifies the immediate kernel problem, while the JL bound supplies a principled benchmark for deciding whether a chosen representation is dimensionally adequate.

Care is required around the word *necessary*. JL is primarily a sufficient dimension-reduction theorem. It does not, by itself, prove that semantic content is intrinsically

high-dimensional, that every dataset needs $O(\varepsilon^{-2} \log m)$ coordinates, or that no specially structured political dataset can be represented faithfully on a line. Some configurations may have low intrinsic dimension. However, lower-bound work shows that the JL dependence is essentially optimal for linear dimensionality reduction in worst-case regimes (Larsen and Nelson, 2016). The qualified conclusion is therefore a geometry-preserving necessity: if one seeks a uniform low-distortion representation for arbitrary sufficiently rich finite configurations using linear maps, a target dimension on the JL scale can be unavoidable. One dimension cannot be presumed adequate before the configuration's structure is known.

This qualification strengthens rather than weakens the critique of the centered political axis. The problem is not that one dimension is always forbidden. It is that the one-dimensional taxonomy is chosen before article-conditioned geometry is measured, and its losses are rarely expressed as a distortion budget. High-dimensional embeddings are methodologically necessary for this project in the practical, hypothesis-preserving sense that they supply room for potentially independent semantic relations to remain distinguishable until evidence warrants reduction. Dimensionality reduction can then occur downstream, with the projection, its kernel, and its observed distortions available for inspection.

PCA again offers a helpful contrast. It chooses directions that maximize variance in a supplied dataset, but high variance is not automatically equivalent to political importance, minority protection, factual reliability, or interpretive stability. A low-variance direction may encode the very distinction relevant to a marginalized observer. Similarly, a random JL projection is valuable for distance preservation but does not determine which distances are normatively important. Neither technique delivers neutrality. They make explicit, mathematically testable trade-offs that a flat center too easily conceals.

The vertical approach accordingly delays reification. It begins in a representation dimension capable of retaining multiple observer-conditioned relations, measures pairwise and local structure, and only then produces views suited to human inspection. Each view remains a shadow, but the architecture records which light cast it. Where a projection

preserves neighborhoods, that stability is evidence. Where it merges distinct paths or changes the nearest neighbors of an interpretation, the failure is part of the result rather than noise swept toward the center.

1.9 Contribution and transition to methodology

The argument can now be summarized as a mismatch of units of analysis. Ground News presents article clusters while assigning their politically salient bias coordinates at the publisher level. The rating is useful exogenous context, but it is not a semantic measurement of the article–hypothesis relation. Upstream survey perspectives are condensed into source metadata, and that metadata is then rendered through a centered one-dimensional taxonomy. The system inherits the CrowdTruth intuition that multiple perspectives can be better than one, but it stops before making the geometry among those perspectives an explicit object of inquiry.

This paper's contribution is to formulate the alternative vertically. It retains the original project's multistage zero-shot NLI scope, uses complex embeddings to study the general shape of bias and its component physical properties, and introduces observer-indexed metrics and relations rather than a single unmarked axis. It is critical of both pure relativism and agreement imposed for its own sake. Goodman and Kuhn explain why measurement participates in world constitution; Bowker and Star explain why the resulting classifications acquire infrastructural force; Wittgenstein warns against placing the describing standpoint inside its own map as though it were neutral. Perspectivist NLP and CrowdTruth show how disagreement can become data, while journalistic ethics and democratic theory prevent that insight from deteriorating into factual indifference.

The topos-theoretic and HoTT-informed framing supplies a language for local sections, restrictions, transport, paths, and possible gluing without pretending that these concepts implement themselves. The JL lemma supplies a quantitative reason to preserve adequate dimensionality before reducing a semantic configuration to an interface. Together they relocate neutrality from an assumed coordinate to a question that can be tested: does a candidate relation remain stable under relevant changes of observer,

metric, hypothesis, and projection?

The following methodology addresses three forms of concealment identified in the source draft: an ontological discrepancy between the object displayed and the object measured; perspectival opacity about who or what supplies the coordinate frame; and concatenation, in which successive classifications hide the values that constitute discourse. It develops the vertical tracks by which semantic dimensions and local physical properties are estimated and by which movement across the learned manifold is used to probe perspectival boundaries. The objective is not to abolish centers, classifications, or projections. It is to stop mistaking their shadows for a view from nowhere.

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